

Contents

Introduction	vii
1 Some infinitary combinatorics	1
Notes for Chapter 1	17
2 Introducing the chain conditions	21
Chain conditions	21
The λ -Souslin numbers	29
Reduction to compact Hausdorff spaces	34
Notes for Chapter 2	40
3 Chain conditions in products	43
Product theorems for regular cardinals	48
Product theorems for singular cardinals	53
Chain conditions in powers	61
The numbers $S_\lambda((X_I)_\kappa)$	69
Notes for Chapter 3	75
4 Classes of calibres, using Σ-products	78
Notes for Chapter 4	84
5 Calibres of compact spaces	86
Strongly $S(X)$ -inaccessible cardinals as calibres	87
(Pre-) calibres for P_{λ^+} -spaces and λ^+ -box products	96
Calibres of compact spaces: the 'exceptional' case	105
Examples on calibres	111
Notes for Chapter 5	116
On the existence of independent families	118
On the isomorphic theory of Banach spaces	121

6	Strictly positive measures	124
	Characterization of spaces with a strictly positive measure	125
	Spaces with a strictly positive measure	133
	The properties $K_{\alpha,n}$	135
	Gaifman's example	147
	The example of Argyros	151
	The Galvin–Hajnal example	162
	Property (**) for non-compact spaces	167
	Notes for Chapter 6	172
7	Between property (K) and the countable chain condition	180
	Kunen's example	180
	The Laver–Galvin example	190
	Notes for Chapter 7	200
	Corson-compact spaces	204
8	Classes of compact-calibres, using spaces of ultrafilters	208
	Notes for Chapter 8	217
9	Pseudo-compactness numbers: examples	218
	Notes for Chapter 9	229
10	Continuous functions on product spaces	231
	Notes for Chapter 10	247
	Appendix: preliminaries	251
	A Set theory and cardinal arithmetic	251
	B Topology and Boolean algebras	260
	C Measure theory	273
	References	281
	Subject index	295
	Index of symbols	300